Physical Sciences for Grade 10

A short test textbook for the HighSchoolEcosystem AI import

Sample material written for software testing only.

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Chapter 1: Matter and its classification

Everything around you that has mass and takes up space is matter. Scientists classify matter so that its behaviour can be predicted and explained. The most basic division is between pure substances and mixtures.

1.1 Pure substances

A pure substance has a fixed composition and definite properties, such as a fixed melting point and boiling point. Pure substances are either elements or compounds. An element is a substance made of only one type of atom, for example oxygen (O) or iron (Fe). Elements are listed on the periodic table.

A compound is a substance made of two or more different elements chemically bonded in a fixed ratio. Water (H_2O) always contains two hydrogen atoms for every oxygen atom. A compound can only be separated into its elements by a chemical reaction, not by physical methods.

1.2 Mixtures

A mixture contains two or more substances that are not chemically bonded. Its composition can vary, and its components keep their own properties. A homogeneous mixture has the same composition throughout, such as salt dissolved in water. A heterogeneous mixture has visibly different parts, such as sand in water.

1.3 Separating mixtures

Because the parts of a mixture are not bonded, they can be separated by physical methods. Filtration separates an insoluble solid from a liquid. Evaporation recovers a dissolved solid by removing the liquid. Distillation separates liquids with different boiling points. Chromatography separates substances that travel at different speeds through a medium such as paper.

1.4 Metals, non-metals and metalloids

Elements can also be classified by their properties. Metals are usually shiny, malleable and good conductors of heat and electricity. Non-metals are usually dull, brittle as solids and poor conductors. Metalloids, such as silicon, have properties between the two and are important in electronics.

Worked example: A student heats a clear liquid. It boils at exactly 100 °C at sea level and leaves no residue. This fixed boiling point suggests the liquid is a pure substance, most likely pure water.

Chapter 2: States of matter and the kinetic molecular theory

Matter exists in three common states: solid, liquid and gas. The kinetic molecular theory explains these states in terms of the particles that make up matter.

2.1 The kinetic molecular theory

The theory states that all matter is made of tiny particles that are always moving. The particles have spaces between them and attract each other. The average kinetic energy of the particles increases as temperature increases.

In a solid, particles vibrate about fixed positions and are held closely by strong forces, so a solid has a fixed shape and volume. In a liquid, particles are close together but can slide past one another, so a liquid has a fixed volume but takes the shape of its container. In a gas, particles are far apart and move freely and quickly, so a gas has no fixed shape or volume.

2.2 Changes of state

Adding energy can make particles move enough to overcome the forces between them. Melting changes a solid to a liquid, and boiling or evaporation changes a liquid to a gas. Removing energy reverses these: condensation changes a gas to a liquid, and freezing changes a liquid to a solid. Sublimation is the direct change from solid to gas, as happens with dry ice.

During a change of state the temperature stays constant even though energy is still being added. The energy is used to overcome the forces between particles rather than to increase their kinetic energy. This is why a heating curve has flat sections at the melting and boiling points.

2.3 Diffusion

Diffusion is the movement of particles from a region of high concentration to a region of low concentration. It happens fastest in gases, slower in liquids, and extremely slowly in solids, because gas particles move fastest and have the most space between them.

Chapter 3: Motion in one dimension

To describe motion we need a reference point and a direction. Motion in one dimension means motion along a straight line, such as a car on a straight road.

3.1 Distance and displacement

Distance is the total length of the path travelled. It is a scalar quantity: it has magnitude only. Displacement is the change in position from the start point to the end point, in a straight line. It is a vector quantity: it has magnitude and direction. A learner who walks 40 m east and then 10 m west has travelled a distance of 50 m but has a displacement of 30 m east.

3.2 Speed and velocity

Average speed is total distance divided by total time. Average velocity is displacement divided by time, and includes a direction. Both are measured in metres per second ($m \cdot s^{-1}$).

3.3 Acceleration

Acceleration is the rate of change of velocity: $a = (v_f - v_i) / \Delta t$, measured in $m \cdot s^{-2}$. An object that speeds up in the direction of motion has positive acceleration in that direction. An object that slows down is accelerating opposite to its motion.

Worked example: A taxi pulls away from rest and reaches 15 $m \cdot s^{-1}$ east in 5 s. Its acceleration is $(15 - 0) / 5 = 3$ $m \cdot s^{-2}$ east.

3.4 Graphs of motion

On a position–time graph, the gradient gives the velocity. On a velocity–time graph, the gradient gives the acceleration and the area under the graph gives the displacement. A horizontal line on a velocity–time graph means constant velocity, so zero acceleration.

Glossary

Compound: Two or more elements chemically bonded in a fixed ratio.

Diffusion: Movement of particles from high to low concentration.

Displacement: Change in position, with direction.

Element: A substance made of one type of atom.

Mixture: Substances combined without chemical bonding.

Vector: A quantity with magnitude and direction.